



PROJECT PLAN

Centralized Agenda

Version 0.4



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Version Control			
Version	Author	Date	Changes
0.1	Iarina Dicu	04-02-2026	Make the layout of the project plan
0.2	Kyan Jeuring	05-02-2026	Updated layout of the project plan
0.3	Kyan Jeuring	05-02-2026	Added chapter 2 Scope and Boundaries
0.4	Iarina Dicu	05-02-2026	Doing Introduction chapter
0.5	David Corodeanu	13-02-2026	Added chapter 5 and 6

Table 1: Version Control

Table of Abbreviations	
Abbreviation	Description
SME	Small and Medium-sized Enterprises
PoC	Proof of concept

Table 2: Table of Abbreviations

1. Introduction

This project focuses on exploring a centralized activity agenda for innovating-related events in the Northern Netherlands. The initiative is proposed by IT-HUB and aims to address the current difficulty of finding relevant events due to information being spread across many different platforms.

At present, hubs, clubs, educational institutions and other organizations each publish their activities in their own way. As a result, businesses and other stakeholders must search multiple websites and calendars to stay informed. This project investigates how these activities could be brought together into one shared agenda to create a clearer and more accessible overview.

The project is carried out as an initial research phase. The main focus is on technical feasibility and validation, rather than on delivering a fully developed or production-ready system.

1.1. The Client: IT-HUB

IT-HUB is an innovation community based in the Northern Netherlands that connects businesses, education, talent, and technology. Its main goal is to support digital innovation and collaboration between organizations, students, and professionals.

The organization works closely with small and medium-sized enterprises (SMEs), educational institutions, and public partners. IT-HUB organizes events, knowledge sessions, and collaborative activities focused on digital topics such as IT, innovation, and technology.

As the main stakeholder of this project, IT-HUB aims to improve how innovation-related activities are shared and discovered across the region, strengthening cooperation between hubs, clubs, and other organizations.

1.2. The Problem

The main challenge addressed by this project is the fragmentation of event information. Innovation-related activities are published on many different platforms, each with its own structure and schedule.

For businesses and other interested parties, this makes it difficult to gain a complete overview of available events. Keeping track of multiple calendars is time-consuming and often leads to missed opportunities. As the number of activities continues to grow, the lack of a centralized overview becomes an increasing problem.

1.3. Project Goal

The goal of this project is to create a centralized activity agenda that brings events from different organizations together in one place.

This agenda helps students and the general public easily find and follow events organized by companies collaborating with IT-HUB. By showing all events on one platform, the project increases visibility and awareness and makes it easier for people to participate.

1.4. Objective

The objective of this project is to deliver a technically validated proof of concept that shows how activities from different hubs and clubs can be combined into one agenda.

This includes researching possible integration methods, such as connectors or APIs, and demonstrating how event data can be collected without requiring changes to the existing systems of participating organizations.

1.5. Expected Results and Success Criteria

The expected outcome of this project is a proof of concept that successfully aggregates event data from a selected number of organizations in the Northern Netherlands.

The project will be considered successful if:

- A suitable technical approach is identified
- Event data can be collected and combined in a reliable way
- Stakeholders gain a clear understanding of how the solution could be expanded in a later phase

Delivering a complete platform or final user interface is outside the scope of this phase.

2. Scope and Boundaries

This chapter defines the scope, boundaries and preconditions of the project Centralized Activity Agenda.

2.1. Purpose

The purpose of this project is to investigate and propose a technical solution for a centralized activity agenda that provides a clear and complete overview of innovation-related activities and events offered by hubs, clubs and educational institutions in the Northern Netherlands.

Currently, activities are published across multiple platforms, making it difficult for SMEs¹ and other stakeholders to find relevant events. This project aims to address that fragmentation by researching and validating a sustainable, maintainable, and scalable way to collect and aggregate event data into one shared agenda.

In this first phase, the focus is on research and proof of concept, not on delivering a fully operational endproduct.

2.2. In Scope

The following elements are included in the scope of the project.

- Research into technical approaches for collecting event data, such as:
 - APIs
 - Connectors
 - Structured data feeds
 - Controlled web scrapping
- Analysis of feasibility, stability, maintainability, and scalability of these approaches.
- A proof of concept demonstrating how event data from a limited number of hubs or clubs can be integrated into a centralized activity agenda
- Stakeholders and events are located in:
 - Groningen
 - Drenthe
 - Friesland

¹ SME is an abbreviation for Small and Medium-sized Enterprises

2.3. Out of Scope

The following are explicitly excluded from the project:

- Full implementation of a production-ready platform
- A complete UI/UX design or finished website
- Manual data entry or long-term operational maintenance
- Events outside the Northern Netherlands (e.g. Amsterdam, Rotterdam)
- Financial monetization models
- Large-scale AI functionality beyond conceptual exploration

2.4. Boundaries

Geographical boundaries

- The project is limited to the Northern Netherlands, specifically the provinces of:
 - Groningen
 - Drenthe
 - Friesland
- National or international events and organizations fall outside the project boundary.

Technical boundaries

- The project focuses on research and validation, not full software delivery.
- Only a limited number of hubs and clubs will be used for testing the proof of concept.
- Existing systems of hubs and clubs will not be modified.
- The solution must be reusable but does not need to be fully integrated during this phase.

Functional boundaries

- The centralized agenda is intended as an overview and redirection tool, not a registration system.
- Users may be redirected to the original event provider for detailed information or sign-up.
- The agenda will not replace existing calendars but complement them.

2.5. Preconditions

The following conditions must be met before and during project execution:

- Access to a list of relevant hubs, clubs, and educational organizations in the Northern Netherlands.
- Willingness of selected organizations to share event data or allow technical inspection
- Availability of stakeholders for weekly feedback moments (especially on Mondays)
- Agreement that this phase focuses on research and proof of concept, not a final product
- Compliance with basic GDPR principles when handling publicly available event data
- Clear communication channels between students and stakeholders

3. Quality Control

Quality control ensures that deliverables meet the agreed requirements through standards, collaboration, and feedback. It is safeguarded by explicit criteria, regular checks, reviews, and testing.

Quality Control		
Standards	Methods	Techniques
NHL Stenden Emmen Coding Conventions (NHL-Stenden-Emmen, 2023)	Peer reviews, continuous integration	Code review, documentation review, unit/integration testing
The 2020 Scrum Guide (Scrumguides.org, 2020)	Definition of Done (DoD)	Issue tracking (JIRA)
Version Control Nottingham Guide (Nottingham University, 2015)	Retrospectives	Version control (GitHub, Git)
APA7 (American Psychological Association, 2020)	Documentation Review	Peer proofreading

Table 3: Quality Control: Standards, Methods and Techniques

4. Project Organization

This chapter overviews how the team is structured, its communication and availability. It describes the roles and responsibilities within the team and clarifies the availability to support effective collaboration during the project.

4.1. Team Structure

This table provides an overview of the team structure and its members roles, contact information as well as the supervising teachers and stakeholders' details.

Team Structure				
Name	Role	Function	Email	Phone Number
Mr. Ranjbar	Product Owner	Owns the product vision and backlog, prioritises features and clarifies the requirements for the team.	amir.ranjbar.maki@student.nhls-tenden.com	0647760294
Mr. Corodeanu	Scrum Master	Facilitates Scrum events and guide the team to follow the correct practices.	david.corodeanu@student.nhlstenden.com	615293706
Mr. Mohr	Developer	Responsible for the development of the project and features implementation.	bence.mohr@student.nhlstenden.com	0616636599
Ms. Dicu	Developer		iarina.dicu@student.nhlstenden.com	0644785667
Mr. Jeuring	Developer		kyan.jeuring@student.nhlstenden.com	0657818340
Mr. Cardoso	Developer		rodrigo.neves.cardoso@student.nhlstenden.com	00617679669
Mr. Gels	Stakeholder	Provide the requirements for the project, feedback and project direction.	e.gels@it-hub.nl	
Ms. Gijsman	Stakeholder		j.gijsman@it-hub.nl	
Mr. Eissens	Stakeholder		r.eissens@it-hub.nl	0624221218
Ms. Zwiers	Stakeholder		h.zwiers@it-hub.nl	
Mr. Meems	Stakeholder		h.meems@it-hub.nl	
Mr. Laan	Teacher	Monitor the team's progress, provide feedback and assess the project outcomes.	rene.laan@nhlstenden.com	0622472058
Mr. Pomp	Teacher		martijn.pomp@nhlstenden.com	0682803986

Table 4: Team Structure

4.2. Communication Plan

- Online client meetings via Microsoft Teams every Monday and in-person client meetings every 2 weeks
- Weekly reports: Stand up meetings Mondays after the client meeting and Fridays
- Urgent issues: via WhatsApp
- Formal updates: via email

4.3. Team Availability

The team will be available for contact Monday to Friday from 9:00 until 17:00 via email WhatsApp or Microsoft Teams.

5. Project Activities

This chapter describes the main technical activities required to develop the proof of concept for the Centralized Activity Agenda during weeks 3 to 8 of the project.

After completing the documentation and research phase in weeks 1 and 2, the project enters the realization phase. During this phase, the focus shifts from analysis to implementation and validation. The objective is to build a technically validated proof of concept consisting of:

- A standardized event data model
- A central aggregation API
- A WordPress plugin that exposes structured event data
- A pull-based synchronization mechanism
- A minimal public website that displays aggregated events

The activities are structured logically and include dependencies to ensure that the system is developed in a stable and controlled manner. Core backend components are prioritized before frontend implementation to reduce integration risks and maintain architectural consistency.

The following table presents the complete list of activities, their purpose, and their dependencies.

Activity table			
ID	Activity	Description	Depends On
A1	Finalize Unified Event Data Model	Define standardized event schema based on requirements and schema.org principles	-
A2	Design System Architecture	Define overall system structure (WP Plugin + Central API + Database + Frontend + Pull logic)	A1
A3	Design Database Schema	Design minimal database model (hubs, events, pull logs) and ERD	A1
A4	Setup Docker Development Environment	Configure backend, database and frontend containers for development	-
A5	Implement Database Setup	Create database, migrations, and initial seed/config for test hubs	A3, A4
A6	Design Central Aggregation API	Define endpoints, validation rules, authentication approach and data flow	A2
A7	Implement Central API Core	Develop API core logic and connect it to the database	A5, A6
A8	Implement Validation & Normalization Layer	Implement schema validation, required fields checks, and duplicate detection logic	A7
A9	Design WordPress Plugin Architecture	Define plugin structure, endpoint format, authentication method and mapping from WP event source	A1, A2
A10	Implement WordPress Plugin	Build plugin that exposes standardized event endpoint (read-only output for pulling)	A9
A11	Implement Pull Mechanism	Build scheduled pulling + sync logic (fetch, compare, upsert, log results)	A7, A10
A12	Implement Logging & Error Handling	Add pull logs, failure reasons, retry strategy and API error responses	A11

A13	Design Public Website Interface	Define wireframes and structure for event listing, filters and detail view	A2
A14	Implement Frontend Website	Build PoC website interface for events	A7, A13
A15	Integrate Frontend with Central API	Connect frontend to API endpoints and implement filtering/search display	A14, A7
A16	Implement Basic Security Layer	Add API keys for hubs/endpoints, input sanitization, basic rate limiting	A7
A17	System Integration Testing	Test full flow: WP Plugin → Pull → API → DB → Website (+ non-WP API path)	A11, A15, A16
A18	Validation with Selected Hubs	Validate using 2–3 real or simulated hub endpoints and record outcomes	A17
A19	Technical Documentation	Document schema, database, API endpoints, plugin usage, pull strategy and setup	A17
A20	Demo Preparation & Final Validation	Prepare demo scenario and final stakeholder-ready explanation	A18, A19

Table 5: Table of Activities

The activity structure reflects a backend-first development strategy. The unified event data model and system architecture form the foundation of the project. Based on this foundation, the database and central aggregation API are implemented.

In parallel, the WordPress plugin is developed to provide standardized event endpoints. Once both components are available, the pull mechanism is implemented to synchronize event data into the central system.

The frontend website is developed after the API core is stable to ensure reliable integration. Security, validation, and logging mechanisms are added to improve robustness and demonstrate technical feasibility.

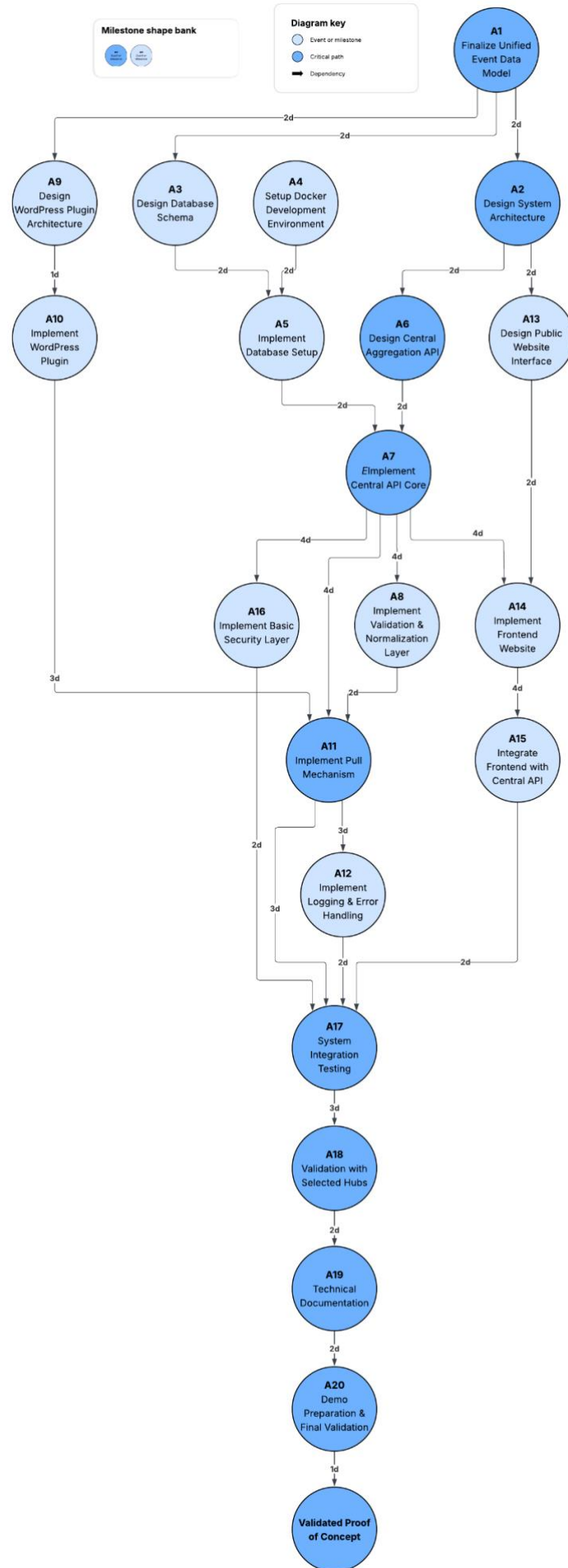
The final phase focuses on integration testing, validation with selected hubs, documentation, and preparation of the final demonstration.

This structured approach ensures that the proof of concept remains aligned with the project scope and delivers a stable and demonstrable solution within the six-week realization period.

Appendix A – Project Activity Diagram

The activity diagram below (next page) is presented in a reduced format for inclusion in the main document.

The original layout and full-scale version can be found in Appendix A for detailed inspection.



6. Planning

This chapter presents the time planning for the realization phase of the Centralized Activity Agenda proof of concept. Following the completion of the documentation phase in weeks 1 and 2, the project enters a six-week development period (weeks 3–8).

The planning is based on the activity structure defined in Chapter 5. Dependencies between activities are respected, and parallel work is applied where possible to ensure completion within the available timeframe.

6.1. Planning Approach

The planning follows a backend-first development strategy. Core architectural and technical components are implemented before integration and validation phases begin.

The following principles guided the planning:

- The unified data model and architecture are finalized before implementation.
- The central API and database are prioritized.
- The WordPress plugin is developed in parallel with backend components.
- The frontend is implemented after API stability is ensured.
- Integration testing and validation conclude the development phase.

The total realization period spans 30 working days.

6.2. Gantt chart

The detailed Gantt chart corresponding to this planning is provided in Appendix B.

7. Sources / References

All sources are formatted according to APA guidelines including full date access and source details.

IT-HUB. (n.d.). *About IT-HUB*. <https://it-hub.nl/over-ons/>